

Integrative Transcriptomic and Machine Learning Approaches for Biomarker Discovery and Fairness Assessment in Lung Cancer

Submitted by
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DECLARATION

I hereby declare that the dissertation entitled “**Integrative Transcriptomic and Machine Learning Approaches for Biomarker Discovery and Fairness Assessment in Lung Cancer**” is submitted for partial fulfilment for completion of my project for **Master of Science in Bioinformatics**. This work is entirely an original done by me at the **Department of Bioinformatics**, Pondicherry University and Pondicherry. This work has not been submitted elsewhere for any other degree, diploma, associateship or fellowship.

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1. ABSTRACT

Lung Adenocarcinoma (LUAD) continues to be a major cancer-related death factor around the world, which demonstrates the necessity of dependable molecular biomarkers to improve diagnostic and treatment methodologies. This study analyzed RNA-Seq gene expression profiles and clinical information from 600 LUAD samples identifying 2,269 significant Differentially Expressed Genes (DEGs). The DEGs mainly enriched in cell cycle regulation, extracellular matrix organization, and cytokine-mediated signaling pathways. Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways such as neuroactive ligand-receptor interactions, hormone signaling, and metabolic processes. Through Protein-Protein Interaction (PPI) network construction and hub gene analysis, three central regulatory genes, CCNB1, CENPA, and TOP2A, were identified. Survival analysis revealed that high expression of these genes correlates with poorer patient prognosis. Furthermore, Receiver Operating Characteristic (ROC) curve analysis yielded Area Under Curve (AUC) values >0.95 , indicating potential diagnostic biomarkers. Stage specific expression profiling confirmed progressive upregulation of these hubs across tumor stages, suggesting their role in disease progression. Finally, evaluation of a fairness-aware machine learning classifier trained on these hub genes ensured equitable prediction performance highlighting potential for unbiased clinical application.

2. INTRODUCTION

Lung cancer continues to be one of the most destructive and lethal cancers globally. It is responsible for approximately one of five cancer-related deaths worldwide, which is a significant public health problem [1]. Among the various histological categories of lung adenocarcinoma (LUAD) is the most common subtype, accounting for 40 – 50% of non-small cell lung cancer (NSCLC) [3,2]. Although much clinical research has been carried out in recent decades, LUAD is frequently diagnosed at a late stage with vague or even no early symptoms, which has led to poor survival and a high recurrence rate [4].

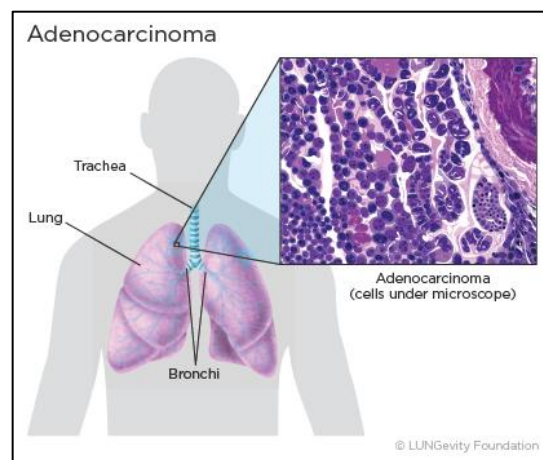


Figure 2.1: Anatomical diagram of lung adenocarcinoma (Source: <https://www.lungevity.org>)

Conventional diagnostic strategies like histopathology or imaging are of vital importance, but not sufficient for early diagnosis or personalized approach planning [5]. The 5-year survival rate of LUAD is less than 20%, partly owing to the fact of late diagnosis and resistance to therapy [6]. This has led the scientific community to look for new biomarkers and precision medicine strategies that change how we diagnose, treat, and monitor lung cancer.

Over the past few years, molecular profiling has revolutionized cancer research. Using techniques such as RNA sequencing (RNA-Seq), researchers can investigate the transcriptomic profile of tumors with unprecedented depth. It facilitates the discovery of differentially expressed genes (DEGs) that are crucially involved in the cancer initiation, development, and metastasis [7,8]. In contrast to classical biomarkers merely reflecting

presence rather than stage, transcriptomic biomarkers allow dynamic and stage-specific perspectives critical for timely interventions and risk assignments [9].

In particular, LUAD is caused by multiple levels of molecular mechanisms, such as cellular cell cycle genes dysregulations, chromosomal instability, DNA repair deficiency, immune escape, and metabolic reprogramming [10]. There are potential biomarkers (mutations in EGFR, KRAS, and ALK) that facilitate personalized treatment in some patients [4]. Nevertheless, these targets apply to some but not all LUAD. This underscores a critical unmet need: developing a broader set of molecular biomarkers independent of, or more generalizable across, mutation status.

Systems biology studies have filled this deficiency imperfectly, showing that gene expression networks and their hub genes are the most connected regulators that control critical cellular activities [11]. These genes are then used as more reliable and informative biomarkers than those identified from DEG analysis alone. For example, CCNB1 (Cyclin B1) is involved in uncontrolled growth, TOP2A in DNA topology and chemotherapy sensitivity, and CENPA, a histone variant, in chromosomal segregation and tumor aggressiveness [12,13,14]. However, extensive integrative studies have not validated these candidates in conjunction with clinical variables such as patient survival and cancer stage.

The emergence of AI and machine learning as a dimension of biomarker research demonstrated great potential in clinical prediction problems, going from imaging to genomics [15]. The understanding that models trained on unbalanced datasets develop algorithmic bias is increasing given their skewed nature due to demographic factors like gender, race, and socioeconomic status[16,17]. Within a healthcare setting, bias extends beyond technical errors because it produces unequal outcomes, which worsen current health disparities[18,19].

Evaluating both model performance metrics and the demographic groups from which the model benefits is essential for the ethical and equitable deployment of AI. Fairness in machine learning requires measuring and reducing prediction bias between different demographic groups. The research adopts this guiding principle by discovering molecular biomarkers in LUAD and incorporating fairness evaluations throughout the model

validation steps. This approach connects biological understanding with ethical AI operations, which computational cancer research seldom addresses together.

The future of precision oncology lies in merging biomedical research with data ethics. Precision medicine requires customized treatment plans based on molecular profiles and fair innovations that benefit people of all genders and racial backgrounds. Integrating fair machine learning algorithms into biomarker discovery represents a valid scientific methodology and an essential ethical requirement [16,17].

Ultimately, this study is driven by a simple yet urgent question: Are there reliable biomarkers for LUAD that stand up to scientific validation and equitable application criteria? We address this by deploying an extensive computational pipeline that combines transcriptomic profiling with survival modeling and fairness-aware machine learning. Our aim extends beyond discovering which genes matter in LUAD to developing diagnostic tools that deliver equitable healthcare outcomes for patients without demographic discrimination.

3. REVIEW OF LITERATURE

3.1 LUAD and NSCLC Overview:

Lung cancer remains one of the most prevalent and fatal malignancies worldwide, with NSCLC accounting for approximately 85% of all lung cancer cases. Among the subtypes of NSCLC, LUAD is the most frequently diagnosed, representing nearly 40–50% of NSCLC cases [3]. LUAD originates from glandular epithelial cells, typically arising in the peripheral regions of the lung and is characterized by a variety of histological and molecular features. Unlike other NSCLC subtypes such as squamous cell carcinoma or large cell carcinoma, LUAD tends to be more common among non-smokers and younger patients, especially women [20]. Advances in genomic profiling have significantly improved the classification of LUAD based on molecular alterations such as mutations in EGFR, KRAS, ALK rearrangements, and other actionable oncogenic drivers, which have also paved the way for targeted therapies (Cancer Genome Atlas Research Network, 2014). The heterogeneity in LUAD not only in morphology but also in genetic architecture makes it a complex disease to treat, but also one rich in opportunities for personalized medicine.

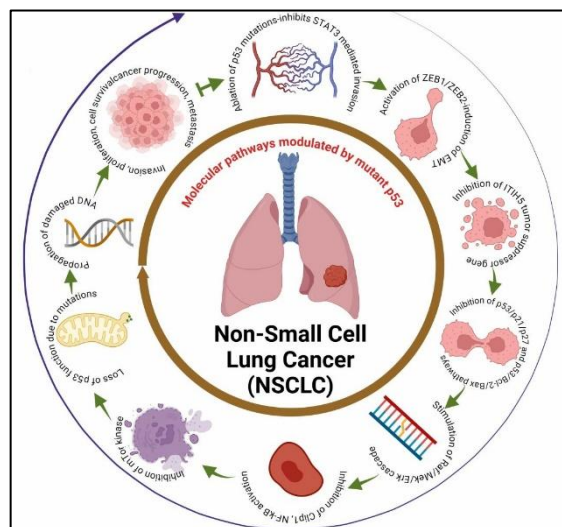


Figure 3.1: Molecular pathways modulated by mutations in Non-Small Cell Lung Cancer (NSCLC), illustrating key genetic alterations and their downstream effects on cellular processes. (Source: Cancer TODAY | IARC – <https://gco.iarc.who.int/today>.)

3.2 Prevalence and Epidemiological Statistics

The global burden of lung cancer remains high, with over 2.2 million new cases and 1.8 million deaths reported in 2020 alone, making it the leading cause of cancer-related mortality [21]. LUAD has shown a steadily increasing incidence, particularly in countries with declining smoking rates, indicating the role of additional environmental and genetic factors. In the United States, LUAD has overtaken squamous cell carcinoma as the predominant histological subtype of lung cancer in both men and women [1]. This shift in histology is also being observed in parts of Asia and Europe, with notable variations influenced by air quality, smoking patterns, and industrial exposure. According to SEER (Surveillance, Epidemiology, and End Results) data, the 5-year survival rate for LUAD remains below 25%, though it is marginally higher than other NSCLC subtypes when diagnosed early (SEER, 2023). Gender-based differences are also evident, with women having a better prognosis and response to certain targeted therapies, possibly due to hormonal or genetic influences [22]. These trends underscore the necessity for early detection strategies and demographic-specific treatment modalities.

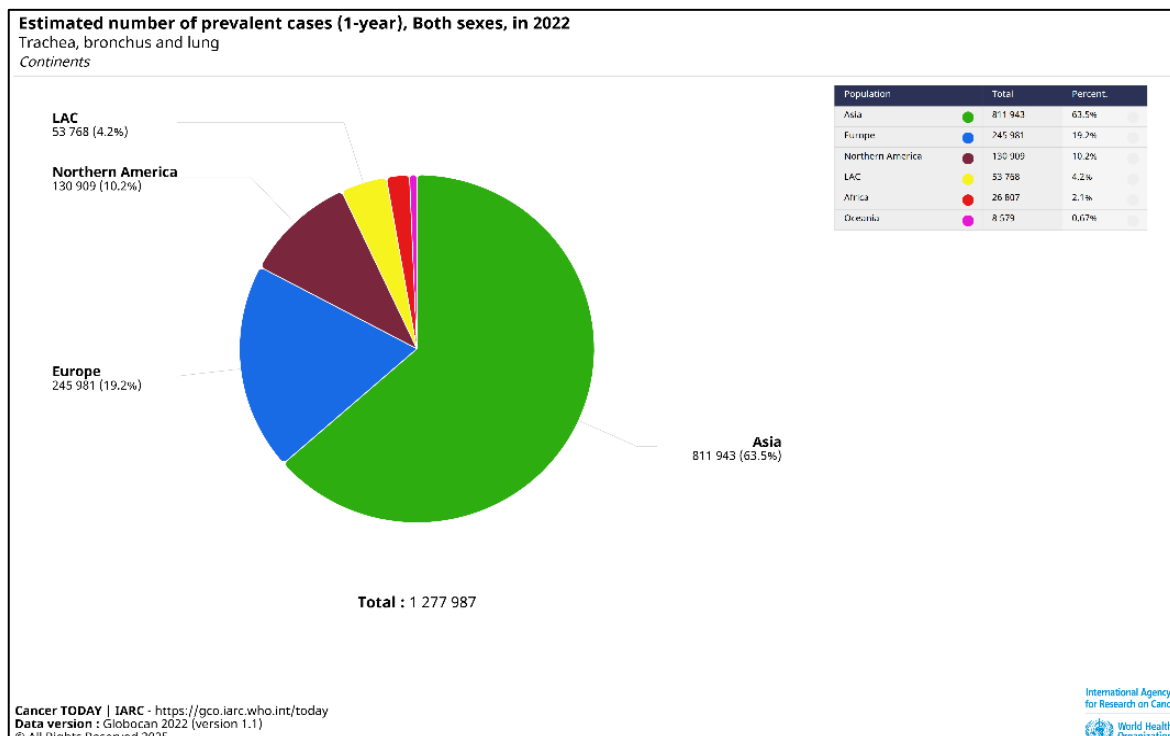


Figure 3.2: Global distribution of respiratory tract cancer cases (trachea, bronchus and lung) by region in 2022, showing Asia with the highest prevalence at 63.5% of worldwide cases. (Source: <https://doi.org/10.1093/nar/gkx1031>)

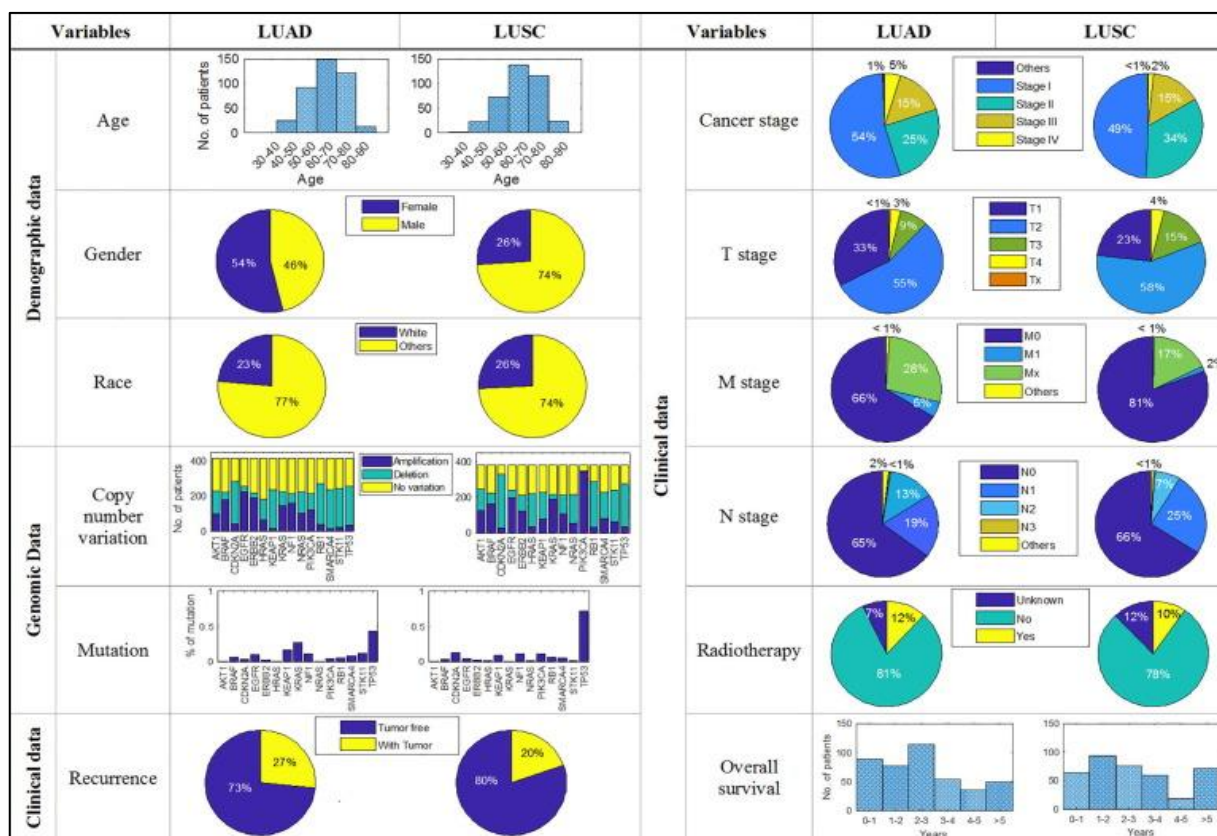


Figure 3.3: LUAD vs LUSC patient comparison: demographics, clinical staging, and survival outcomes.

3.3 Etiological Agents and Risk Factors

The development of LUAD and NSCLC is influenced by a multitude of etiological agents and risk factors that interact in a multifaceted manner. Tobacco smoking remains the predominant risk factor, with over 80% of lung cancer deaths attributable to smoking [23]. However, LUAD is increasingly diagnosed in never-smokers, pointing toward other significant contributors such as exposure to radon gas, air pollutants (especially fine particulate matter PM_{2.5}), occupational exposure to asbestos and heavy metals, and genetic predispositions [24]. Chronic respiratory conditions, such as chronic obstructive pulmonary disease (COPD), also elevate the risk of developing LUAD due to persistent inflammation and epithelial damage. Furthermore, molecular epidemiological studies have implicated germline mutations and single nucleotide polymorphisms (SNPs) in genes such as TP53, EGFR, and BRCA2 in increasing susceptibility to lung cancer [25]. Notably, LUAD in non-smokers often involves mutations in the EGFR gene or ALK translocations,

which are infrequent in smoking-related cases and indicate distinct pathophysiological pathways [26]. These divergent mechanisms of tumorigenesis necessitate tailored risk assessment and prevention strategies.

3.4 Causative Agents of LUAD and NSCLC

Among the causative agents implicated in LUAD and NSCLC, tobacco smoke remains the most potent and extensively studied, containing over 60 known carcinogens that can induce mutations in critical oncogenes and tumor suppressor genes such as *KRAS*, *TP53*, and *EGFR* [27]. Polycyclic aromatic hydrocarbons (PAHs) and tobacco-specific nitrosamines (e.g., NNK) are among the major carcinogens in cigarette smoke that contribute to DNA adduct formation and genomic instability. Radon gas, a naturally occurring radioactive substance, is the second leading cause of lung cancer globally and contributes significantly to disease burden in regions with high geological radon exposure [24]. Airborne pollutants, especially fine particulate matter (PM_{2.5}), diesel exhaust, and volatile organic compounds (VOCs), have also been confirmed as causative agents by the International Agency for Research on Cancer (IARC), classifying outdoor air pollution as a Group 1 carcinogen for lung cancer [28]. Emerging evidence implicates chronic inflammation and oxidative stress as key mechanisms by which these environmental agents exert carcinogenic effects, promoting epithelial-mesenchymal transition and angiogenesis in the lung microenvironment [29]. Thus, a broad spectrum of chemical, physical, and biological agents collectively contribute to LUAD pathogenesis.

3.5 Role of Protein-Coding Genes in Disease Progression

Protein-coding genes play a pivotal role in the pathogenesis, progression, and therapeutic response of LUAD. Mutations in oncogenes such as *EGFR*, *KRAS*, *ALK*, *BRAF*, and *MET* are among the most frequently observed alterations in LUAD and have been extensively studied as both prognostic biomarkers and therapeutic targets [30]. The *EGFR* gene, for instance, encodes a receptor tyrosine kinase involved in cell proliferation and survival. Mutations in its tyrosine kinase domain lead to uncontrolled cell growth and are more prevalent in non-smoking Asian women with LUAD, making EGFR inhibitors a cornerstone of treatment for this subgroup [31]. Similarly, *KRAS* mutations, often linked with smoking, are associated with poor prognosis and resistance to certain therapies,

though recent advances in KRAS G12C inhibitors offer promising options [32]. Tumor suppressor genes like *TP53*, *STK11*, and *KEAP1* are also frequently inactivated in LUAD, contributing to genomic instability, immune evasion, and metabolic reprogramming [33]. Gene expression studies have identified aberrant activity of genes involved in cell cycle regulation (e.g., *CCNB1*, *CDK1*), DNA repair (*BRCA1*, *RAD51*), and angiogenesis (*VEGFA*), which are often upregulated during disease progression and are indicative of aggressive tumor behavior [34]. Integration of multi-omics data has also highlighted the role of alternative splicing and dysregulated RNA-binding proteins in altering the transcriptomic landscape of LUAD [35]. These findings not only enhance our understanding of LUAD biology but also expand the repertoire of potential molecular targets for drug development.

3.6 The Critical Need for Biomarkers in LUAD

Lung adenocarcinoma (LUAD), the most prevalent histological subtype of non-small cell lung cancer (NSCLC), remains one of the most lethal malignancies globally, accounting for over 40% of all lung cancer cases and contributing significantly to cancer-related mortality [1]. Despite advancements in imaging modalities, surgical interventions, chemotherapy, radiotherapy, and the recent implementation of immunotherapy and targeted therapies, the five-year survival rate for LUAD remains alarmingly low, primarily due to the late-stage diagnosis in the majority of patients. Most LUAD cases are asymptomatic during early stages and are often diagnosed incidentally or only after the disease has progressed to an advanced, often metastatic, stage [36]. The conventional diagnostic modalities, such as low-dose computed tomography (LDCT) and histopathological evaluation, although valuable, are not always sufficient due to limitations including false-positive results, radiation exposure, high costs, and invasiveness of biopsy procedures. As such, there is an urgent and unmet clinical need for the development of non-invasive, reliable, and highly specific biomarkers that can facilitate early detection, monitor disease progression, predict treatment response, and enable real-time surveillance of recurrence in LUAD patients [37].

Biomarkers defined as measurable indicators of biological or pathogenic processes or pharmacologic responses to a therapeutic intervention can play a transformative role in the personalized management of LUAD [38]. They offer a window into the molecular and

cellular alterations that drive carcinogenesis, providing valuable insights into tumor heterogeneity, resistance mechanisms, and prognostic indicators. In recent years, the paradigm of precision oncology has underscored the importance of molecular biomarkers, not just as diagnostic tools but also as therapeutic targets that can inform and refine clinical decision-making. For example, the clinical utility of biomarkers such as EGFR mutations, ALK rearrangements, and PD-L1 expression has been well-documented in guiding the use of tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs) in LUAD therapy [6]. However, these markers are only present in a subset of patients, and resistance mechanisms often emerge, underscoring the need for a broader spectrum of biomarkers derived from genomic, transcriptomic, epigenomic, proteomic, and metabolomic studies. Importantly, the development of such biomarkers also holds the potential to reduce healthcare costs by avoiding ineffective treatments and minimizing adverse events, making precision medicine a sustainable and scalable approach in oncology. In summary, biomarker development in LUAD is not only a research imperative but also a clinical necessity that could significantly alter the trajectory of this devastating disease.

3.7 Recent Advances and Strategies in Biomarker Discovery for LUAD

The past decade has witnessed significant progress in the identification and validation of biomarkers for lung adenocarcinoma, driven by the increasing availability of high-throughput multi-omics technologies, integrated data repositories such as The Cancer Genome Atlas (TCGA) and (Gene Expression Omnibus) GEO, and advanced computational tools including machine learning and systems biology approaches. These developments have enabled a shift from single-gene assays to comprehensive molecular profiling, allowing researchers to uncover complex biological networks and novel disease-associated signatures. One of the most promising recent advances is the identification of a plasma lipid signature-based ratio scoring model (LSRscore) that can detect early-stage LUAD with remarkable accuracy. This model, developed through high-throughput lipidomics, achieved an area under the curve (AUC) of 0.972 in the discovery cohort and 0.968 in the validation cohort, demonstrating its robustness and potential as a non-invasive screening tool [40]. The clinical utility of such a lipidomic biomarker is significant given the ease of sample collection (plasma) and the sensitivity of lipid changes to early tumorigenic events, which often precede morphological or symptomatic alterations.

In addition to lipidomics, epigenetic modifications have emerged as critical layers of regulation in LUAD pathogenesis. DNA methylation, particularly in CpG islands of tumor suppressor genes, has been associated with tumor progression, metastasis, and resistance to therapy. A notable study identified a six-gene DNA methylation signature that distinguishes LUAD from adjacent normal tissue, with strong discriminatory power across multiple patient cohorts [44]. Such methylation biomarkers are stable, detectable in liquid biopsies (e.g., circulating tumor DNA), and can serve both diagnostic and prognostic purposes. Meanwhile, transcriptomic studies continue to reveal the expression dynamics of genes involved in immune regulation and tumor microenvironment remodeling. For example, CYP27A1, a cholesterol-metabolizing enzyme, was found to be significantly downregulated in LUAD, correlating with poor overall survival and altered immune cell infiltration [45]. This points to the gene's dual role as a prognostic marker and a potential therapeutic target, given its involvement in lipid metabolism and immune response.

Furthermore, the integration of artificial intelligence and machine learning into biomarker discovery pipelines has allowed for more nuanced stratification of LUAD patients based on gene expression, mutation burden, immune checkpoint activity, and clinical variables. Predictive models trained on large multi-omics datasets are now capable of identifying composite biomarkers or gene signatures that outperform traditional univariate markers in prognostic accuracy. For instance, deep learning algorithms have been employed to identify radiomic-genomic signatures that correlate with PD-L1 expression and tumor mutation burden, two important predictors of immunotherapy response. The incorporation of such integrative biomarkers into clinical workflows promises not only to improve early detection but also to personalize treatment regimens and monitor therapeutic efficacy in real time. Overall, the expanding repertoire of biomarkers being discovered through genomic, epigenomic, proteomic, metabolomic, and computational approaches underscores a new era in LUAD management one that is molecularly informed, patient-specific, and dynamic.

3.8 Inclusion of Machine Learning in LUAD Diagnosis and Prognosis

The integration of machine learning (ML) techniques into the diagnosis and prognosis of lung adenocarcinoma (LUAD) represents a significant advancement in clinical oncology, aiming to improve both the accuracy and efficiency of patient management. Traditional

diagnostic methods, including imaging and histopathology, are limited by interobserver variability, subjectivity, and the invasive nature of biopsies. Machine learning, with its ability to analyze vast amounts of data, including medical imaging, genomic profiles, and clinical histories, offers a powerful tool to overcome these limitations. A recent study by [44] demonstrated that machine learning algorithms, particularly deep learning models, could effectively differentiate between LUAD and other types of non-small cell lung cancer (NSCLC) using radiomics data extracted from CT images. These models achieved a sensitivity of 92% and specificity of 89%, outperforming traditional radiologists' interpretations [44]. By extracting features from imaging data that are often imperceptible to human observers, machine learning models can detect early-stage tumors, monitor disease progression, and predict patient outcomes with a level of precision that would otherwise be difficult to achieve.

Moreover, the advent of multi-omics data integration has further enhanced the capability of machine learning in LUAD prognosis. ML models trained on genomic, transcriptomic, proteomic, and metabolomic data can uncover complex molecular signatures that correlate with clinical outcomes, such as survival rates, recurrence, and therapeutic responses. For example, a study by Liu et al. [45] used a random forest model to integrate gene expression data with clinical parameters and successfully identified a five-gene signature that predicted overall survival in LUAD patients with an accuracy of 85%. These types of ML models not only assist in prognostic stratification but also have the potential to guide personalized treatment decisions. In the realm of immunotherapy, ML models that analyze immune landscape features (such as tumor mutational burden and PD-L1 expression) can help predict patient responsiveness to immune checkpoint inhibitors, offering a promising approach to precision medicine [43]. Furthermore, machine learning approaches are being developed to predict the emergence of drug resistance in LUAD, a challenge that has traditionally been difficult to address. These models enable clinicians to anticipate potential resistance mutations in real-time and adjust treatment regimens proactively, thus improving patient outcomes and reducing the burden of ineffective therapies.

Incorporating machine learning into LUAD diagnosis and prognosis not only enhances clinical decision-making but also facilitates the development of novel therapeutic targets. By identifying biomarkers linked to specific disease subtypes and patient characteristics, ML models can uncover molecular pathways that may serve as new targets for drug

development. However, while these technologies show tremendous promise, their widespread clinical adoption requires overcoming challenges related to data quality, model interpretability, and validation across diverse patient populations. Ensuring the robustness, generalizability, and clinical utility of machine learning models will be crucial in making them a routine part of LUAD management [46].

3.9 Research Gaps in LUAD and Current Challenges

Despite the remarkable progress made in understanding the molecular landscape of lung adenocarcinoma (LUAD) and the integration of cutting-edge technologies like machine learning, significant research gaps remain that limit the advancement of diagnostic and therapeutic strategies for this disease. One of the primary challenges in LUAD research is the heterogeneity of the tumor itself. LUAD, like many cancers, exhibits a high degree of genetic and phenotypic diversity, both within a single tumor and between different patients, making it difficult to identify universal biomarkers or develop one-size-fits-all therapeutic approaches. This intra-tumor heterogeneity is a major barrier to treatment efficacy, as it leads to varying responses to therapies, including chemotherapy, targeted therapy, and immunotherapy. Moreover, the tumor microenvironment (TME), which consists of stromal cells, immune cells, and extracellular matrix components, plays a crucial role in LUAD progression, metastasis, and resistance to treatment. However, the complex interactions within the TME remain poorly understood, and there is a pressing need for more research into how these interactions influence tumor behavior and therapy resistance [47]. In this context, the discovery of specific biomarkers within the TME that can predict treatment response is still in its infancy, and this represents a major research gap.

Another major area of concern is the late-stage diagnosis of LUAD, which remains the primary reason for its high mortality rate. Although imaging technologies like CT scans and PET scans are invaluable tools, they are often unable to detect small or early-stage tumors with sufficient sensitivity. Furthermore, biopsy procedures, while essential for confirming the diagnosis, are invasive and carry potential risks. Liquid biopsy, which detects circulating tumor DNA (ctDNA), exosomes, or microRNAs in blood samples, holds great promise as a non-invasive diagnostic method, but it is not yet widely implemented in clinical practice due to issues of sensitivity and standardization [48]. To

bridge this gap, more research is needed to identify reliable liquid biopsy biomarkers for early-stage LUAD detection and to refine the technical aspects of liquid biopsy assays.

In addition to these challenges, the therapeutic landscape of LUAD is complicated by the emergence of resistance to targeted therapies and immunotherapies. For instance, while EGFR inhibitors and ALK inhibitors have been proven effective in patients harboring corresponding mutations or fusions, the development of resistance to these therapies is inevitable, often within months of treatment initiation. The mechanisms underlying this resistance, such as secondary mutations or activation of alternative signaling pathways, are not fully understood. Additionally, despite the success of immune checkpoint inhibitors like pembrolizumab and nivolumab in treating certain LUAD patients, only a subset of patients benefits from these therapies. Therefore, a better understanding of the tumor immune microenvironment (TIME) and the identification of novel immune-related biomarkers are crucial to improving patient selection for immunotherapy. Research into combinatorial therapies that target both tumor cells and the immune microenvironment may offer a promising solution to these challenges [4].

Moreover, the current gap in clinical trials designed to incorporate personalized medicine, particularly in the context of advanced genomic profiling and machine learning algorithms, limits the ability to tailor treatments to individual LUAD patients. While large-scale genomic studies, such as The Cancer Genome Atlas (TCGA), have provided invaluable data, translating these findings into actionable clinical decisions remains an ongoing challenge. Multicenter trials that incorporate real-world data (RWD) and real-world evidence (RWE) are essential to ensure that emerging molecular insights and machine learning models are validated across diverse patient populations and healthcare settings. The ultimate goal is to shift from a one-size-fits-all approach to a truly personalized, precision oncology framework that addresses the unique molecular features of each patient's tumor and guides therapeutic decision-making.

In summary, while significant strides have been made in the molecular characterization of LUAD and the integration of machine learning for diagnosis and prognosis, substantial research gaps remain. A deeper understanding of tumor heterogeneity, the tumor microenvironment, and resistance mechanisms, along with advancements in non-invasive diagnostic methods and personalized treatment strategies, is essential for improving patient

outcomes in LUAD. Addressing these gaps will pave the way for more effective treatments and earlier, more accurate diagnoses, ultimately improving survival rates and quality of life for LUAD patients.

Despite the advancements in biomarker discovery and the integration of machine learning for improved diagnosis and prognosis of lung adenocarcinoma (LUAD), a critical and often overlooked dimension in current research is the assessment of algorithmic fairness across demographic subgroups. Existing prognostic models, while demonstrating high accuracy and clinical utility, frequently lack evaluation for equitable performance among diverse populations, including variations by gender, race, and socio-economic status. This limitation raises serious concerns regarding potential bias and healthcare disparities that may arise when deploying such models in real-world clinical settings. Addressing this research gap, the present study incorporates a comprehensive fairness evaluation framework to assess and ensure that the prognostic model performs equitably across different demographic groups. By employing fairness metrics such as demographic parity and equal opportunity, alongside traditional performance metrics, this study not only strengthens the reliability of the predictive models but also aligns with emerging priorities in ethical AI and precision medicine. In doing so, it contributes to the growing discourse on responsible machine learning in oncology and offers a novel direction for future research aimed at developing clinically robust and socially equitable prognostic tools for LUAD.

4. OBJECTIVES

- To identify biomarkers in lung cancer development and progression through integrative analysis of transcriptomics data
- To evaluate the prognostic and diagnostic potential of the identified key genes
- To develop and assess the fairness and performance of machine learning models

5. METHODOLOGY

5.1 Data Collection

The RNA-Seq Level-3 gene expression profiles and clinical information for LUAD were retrieved from TCGA database (<https://portal.gdc.cancer.gov>) using the TCGAbiolinks R package [49]. A total of 600 samples, comprising 59 normal and 541 tumor samples, were included in the study. The selection and integration of TCGA data were further supported using updated TCGAbiolinks functions for harmonized data processing [50].

5.2 Differentially Expressed Gene (DEG) Identification

Differential expression analysis was conducted using the DESeq2 R package [51], an extension of the original DESeq model for RNA-seq data normalization and dispersion estimation [52]. The raw count data were normalized. Genes with an adjusted p-value (P_{adj}) < 0.05 and an absolute log2 fold change ($|\log_2FC|$) ≥ 2 were considered significantly differentially expressed. Gene annotations, including Ensembl IDs, gene symbols, biotypes, and descriptions, were obtained using the biomaRt R package [53]. Only protein-coding genes were retained for further analysis. DEGs were visualized using volcano plots generated by the ggplot2 R package [54].

5.3 Functional and Pathway Enrichment Analysis

Gene Ontology (GO) enrichment analysis covering Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) categories, as well as Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment, was performed using the clusterProfiler R package [55]. Gene annotation was based on the org.Hs.eg.db package [56]. The GO classification is derived from a widely adopted hierarchical structure used to unify biological data [57]. Enrichment results were visualized with ggplot2 better to understand the biological significance of the identified DEGs.

5.4 Protein-Protein Interaction (PPI) Network and Hub Gene Identification

Protein-protein interaction networks were constructed using the STRING database [58]. The DEG gene symbols were input into the STRING web portal, and the resulting interaction network was visualized and analyzed using Cytoscape software [59]. The

STRING app for Cytoscape enhances data integration for visual and quantitative analysis. Hub genes were identified using the cytoHubba plugin based on three centrality measures: Degree Centrality, Edge Percolated Component (EPC), and Maximum Neighborhood Component (MNC). The top-ranked genes from each method were intersected to determine the final hub genes with the help of a venn diagram.

5.5 Survival Analysis

Kaplan-Meier survival analysis was conducted to evaluate the prognostic significance of the identified hub genes (*CCNB1*, *CENPA*, and *TOP2A*) using the KM Plotter tool [60]. Patients were stratified into high and low expression groups based on median gene expression levels, and survival differences were assessed using log-rank tests.

5.6 ROC Curve Analysis

The classification performance of the hub genes (*CCNB1*, *CENPA*, and *TOP2A*) in distinguishing LUAD from standard samples was evaluated using the Receiver Operating Characteristic (ROC) curve analysis. Logistic regression models were built using Python's scikit-learn library [61] with tenfold cross-validation. Class imbalance was addressed by applying the Synthetic Minority Oversampling Technique (SMOTE) [62]. Model performance was assessed by computing accuracy, sensitivity, specificity, Matthews correlation coefficient (MCC), and the area under the ROC curve (AUC). MCC was included as a robust and informative metric for imbalanced datasets.

5.7 Marker Gene Expression Across LUAD Stages

The expression levels of the hub genes across different LUAD stages were examined using TPM-normalized values, log-transformed as $\log_2(\text{TPM}+1)$. Boxplots were generated using ggplot2 to visualize the stage-wise variation and investigate the potential role of these genes in disease progression.

5.8 Fairness Evaluation in Machine Learning

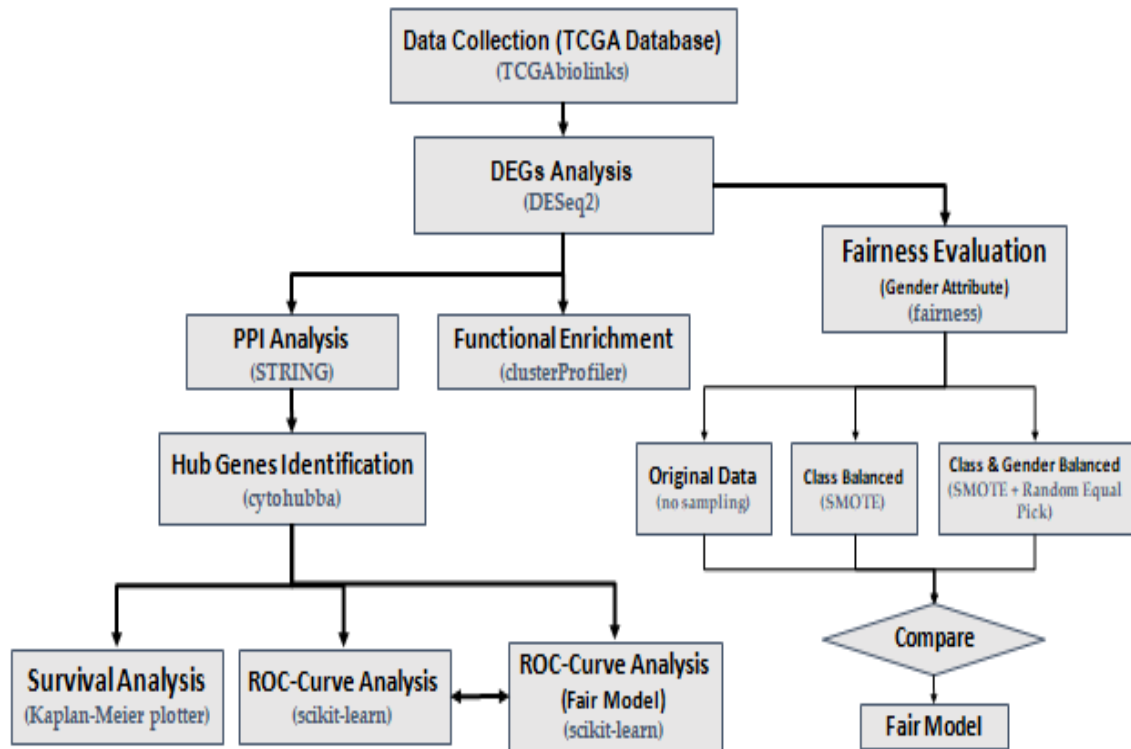
To ensure unbiased predictions in the classification of LUAD samples, a fairness evaluation was conducted with a specific focus on the gender attribute using the fairness R package[63]. The analysis was based on raw count DEG data and aimed to assess the

impact of gender-based bias in model predictions. For this purpose, three versions of the dataset were prepared: the original unbalanced dataset, a class-balanced dataset generated using the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance[62], and a class- and gender-balanced dataset produced by combining SMOTE with random equal sampling across genders. These datasets were used to train machine learning models, which were then evaluated for fairness using metrics such as accuracy and demographic parity. Based on this comparative assessment, the fairest model was selected for downstream performance evaluation.

5.9 ROC Curve Analysis for the Fair Model

ROC curve analysis was then performed on the selected fair model using logistic regression with tenfold cross-validation, implemented in the scikit-learn Python library [61], to assess its classification performance across balanced subgroups (gender) and to compare it with the previous model.

METHODOLOGY



6. RESULTS

6.1 Data Collection and Differentially Expressed Genes (DEGs) Analysis

For this study, RNA sequencing data from 600 lung tissue samples were collected from the TCGA database. These included 59 samples from healthy individuals and 541 from lung adenocarcinoma (LUAD) patients. To identify which genes behaved differently between the healthy and cancerous tissues, DESeq2 tool is used to perform differential gene expression analysis. The analysis revealed a total of 2,269 genes that showed significant changes. Out of these, 579 genes were found to be more active (upregulated) in tumor tissues, while 1,689 genes were less active (downregulated) compared to the normal tissues. These changes were considered significant if they had an adjusted **p-value**<0.05 and **log fold change**>= 2. Volcano plot was created to make it easier to understand which genes were changing and how much (Figure 6.1). In this plot, each dot represents a gene. The red dots show more active genes in cancer, the blue dots show less active genes, and the grey ones don't show significant changes.

These findings show that many genes are turned up or down in lung adenocarcinoma compared to healthy lung tissue. Interestingly, there were far more downregulated genes than upregulated ones. This could mean that many normal functions get suppressed in cancer, possibly including essential protective or tumor-fighting genes. At the same time, the upregulated genes might be playing a role in helping the tumor grow and spread. These differentially expressed genes are crucial, as they give us a starting point to explore what's going wrong at the molecular level in LUAD and which genes might serve as potential targets for diagnosis or treatment.

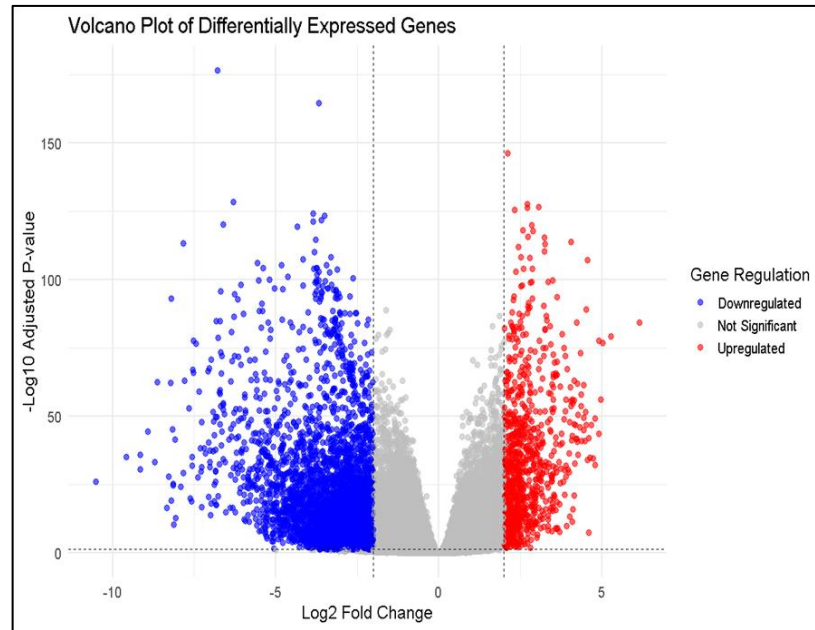


Figure 6.1:Volcano plot of differentially expressed genes (DEGs) in LUAD. Red: Upregulated genes; Blue: Downregulated genes; Grey: Non-significant genes.

6.2 Functional Enrichment Analysis

Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were performed using the clusterProfiler package in R. Functional enrichment analysis was conducted to gain insights into the biological roles and pathways associated with the differentially expressed genes (DEGs) identified in lung adenocarcinoma (LUAD).

The Gene Ontology (GO) analysis revealed that the DEGs were significantly enriched in biological processes related to cell cycle regulation, hormone-mediated signaling, and pattern specification, all critical for tumor development and progression (Figure 6.2). These findings indicate that the dysregulated genes potentially drive uncontrolled cell proliferation and alterations in tissue architecture.

Regarding cellular components, the DEGs showed significant enrichment in the extracellular matrix (ECM) and cell-cell junctions, suggesting a potential role in modifying the tumor microenvironment and promoting metastasis through disrupted intercellular communication and ECM remodelling (Figure 6.3).

The molecular function category highlighted enriched terms associated with peptidase and hydrolase activity, enzymes commonly involved in protein degradation, signaling, and metabolic processes, further supporting a reprogramming of cellular functions in LUAD (Figure 6.4).

KEGG pathway analysis complemented the GO results by identifying several key pathways enriched among the DEGs. Notably, the neuroactive ligand-receptor interaction and adherens junction pathways point to significant alterations in cell signalling and adhesion mechanisms. (Figure 6.5). Enrichment in steroid hormone biosynthesis and arachidonic acid metabolism pathways suggests active metabolic and hormonal changes that may contribute to tumorigenesis and immune modulation.

These enrichment analyses underscore the complex interplay of hormonal regulation, cell communication, and metabolic adaptation in LUAD pathophysiology. The summarized enrichment results are visualized, highlighting the top significant GO terms and KEGG pathways based on adjusted p-values and gene counts.

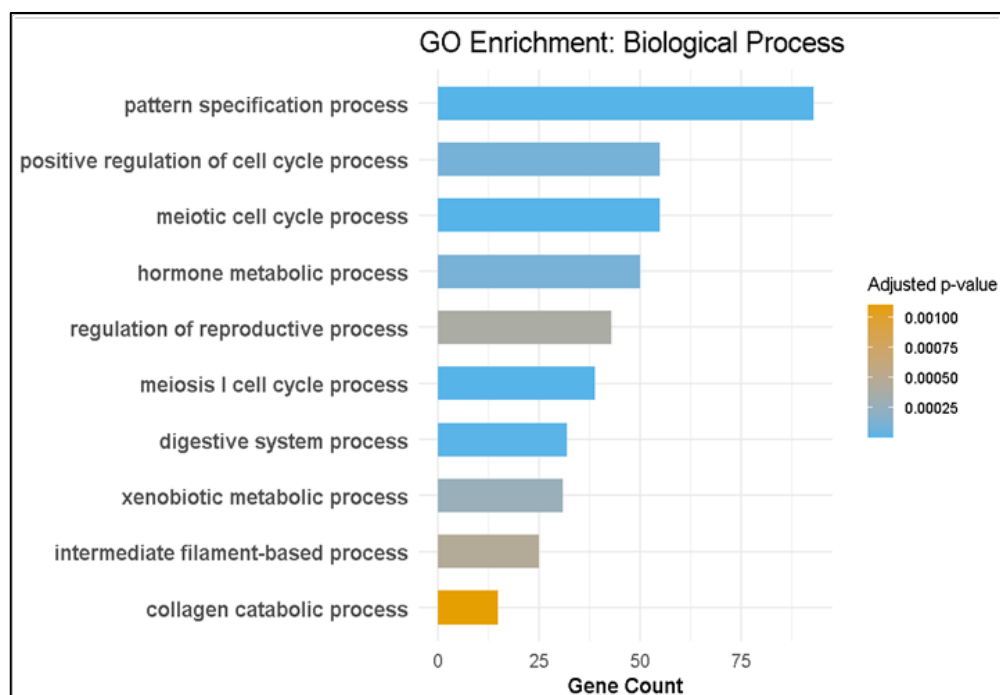


Figure 6.2: Gene Ontology (GO) enrichment analysis of biological processes in lung adenocarcinoma, highlighting dysregulation of pattern specification, cell cycle regulation, and hormone metabolic processes.

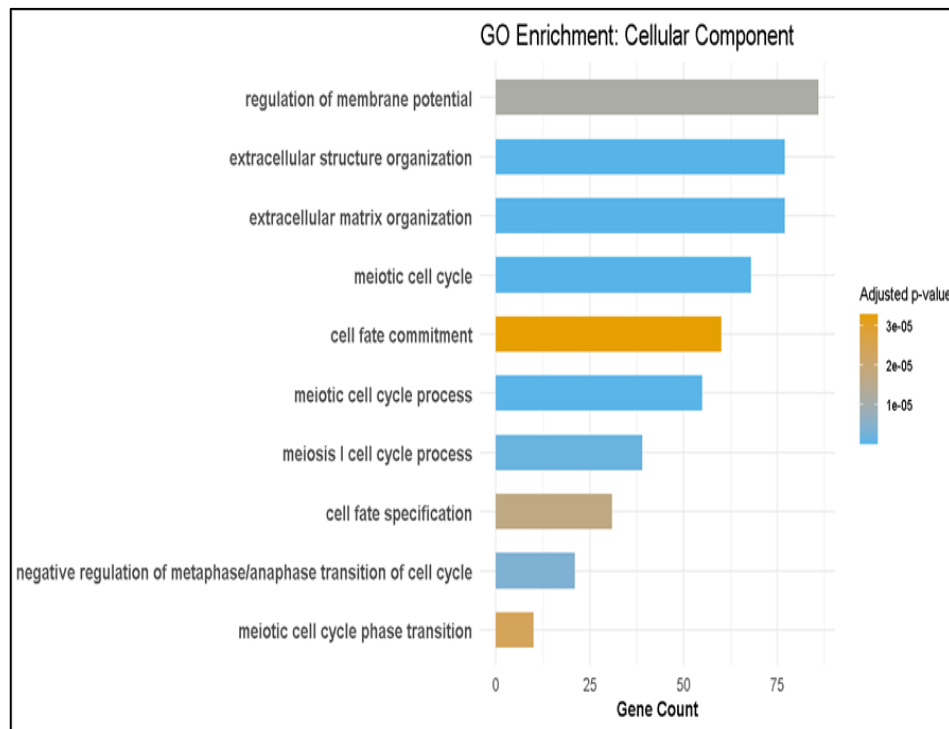


Figure 6.3: Gene Ontology (GO) enrichment analysis of cellular components in lung adenocarcinoma, showing significant alterations in membrane potential regulation, extracellular matrix organization, and cell cycle-related structures.

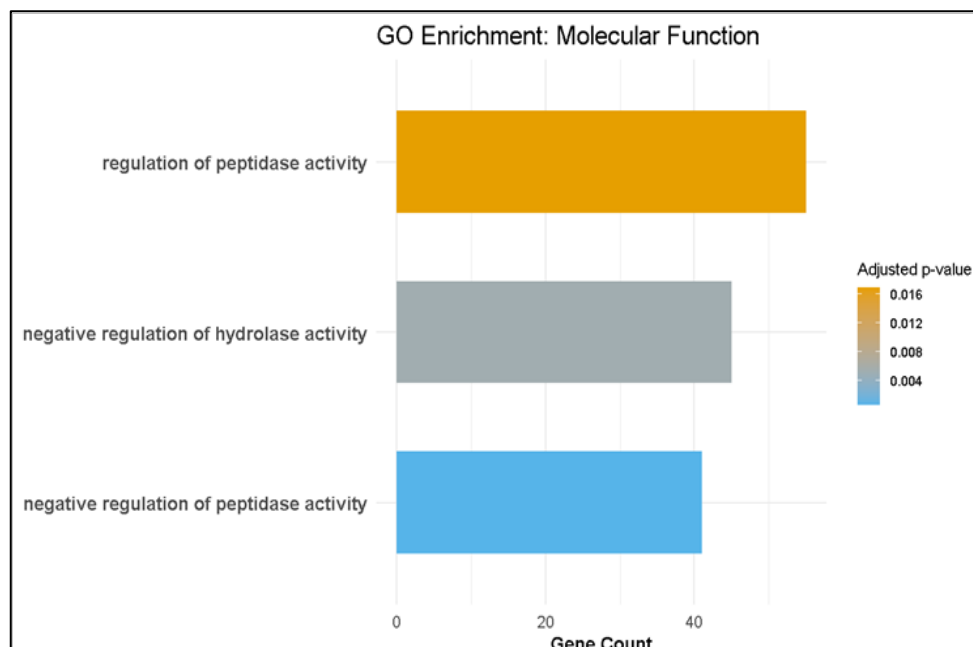


Figure 6.4: Gene Ontology (GO) enrichment analysis of molecular functions in lung adenocarcinoma, demonstrating significant enrichment in peptidase activity regulation and hydrolase activity modulation pathways.

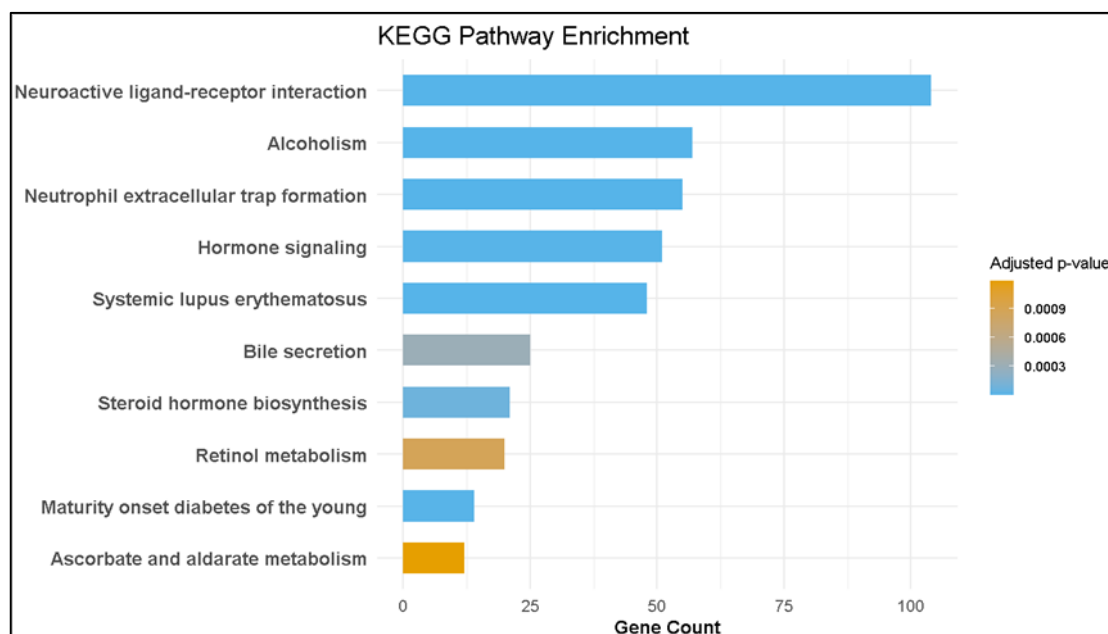


Figure 6.5: KEGG pathway enrichment analysis in lung adenocarcinoma revealing significant involvement of neuroactive ligand-receptor interaction, inflammatory responses, and hormone signaling pathways.

6.3 Protein–Protein Interaction (PPI) Network Construction and Hub Gene Identification

To further investigate the biological significance and interactive behavior of the DEGs identified in LUAD, a PPI network was constructed using data from the STRING database, employing a high-confidence interaction threshold of 0.7 to ensure accuracy and biological relevance. The final PPI network consisted of 2,213 nodes and 6,626 edges, capturing a complex web of functional and physical associations among the DEGs (Figure 6.6). This extensive network structure reflects the multifaceted nature of cellular signaling and regulation in LUAD and underscores the value of interaction-based analyses for uncovering central molecular players within disease-specific networks.

The analysis of such a dense interaction map allows for the identification of hub genes—genes that occupy central and influential positions within the network and are often critical in regulating key cellular functions. To identify these hub genes, the PPI network was analyzed using the CytoHubba plugin within Cytoscape. This tool applies various topological algorithms to determine the importance of individual nodes based on their structural roles in the network. In this study, three distinct algorithms were used to

strengthen the robustness of hub gene identification: Maximum Neighborhood Component (MNC), Degree, and Edge Percolated Component (EPC). These algorithms analyze connectivity and influence from different mathematical perspectives, thereby minimizing potential bias from a single approach.

Across all three algorithms, three genes *CCNB1*, *TOP2A*, and *CENPA* emerged consistently as top-ranking common hub genes (Figure 6.7, Table 6.1) with the help of venn diagram(Figure 6.8) . This strong convergence across methods supports their critical involvement in LUAD pathogenesis. *CCNB1* encodes Cyclin B1, a protein that plays a pivotal role in cell cycle control, particularly in regulating the G2/M transition. Its overexpression has been widely associated with increased proliferation rates and unfavorable prognosis in a range of cancers, including LUAD. *TOP2A* encodes DNA topoisomerase II alpha, an essential enzyme involved in DNA replication, transcription, and chromosomal segregation. This gene is frequently upregulated in tumor cells undergoing rapid division and is a well-recognized target in cancer therapy. *CENPA* encodes a centromere-specific histone H3 variant, essential for kinetochore assembly and accurate chromosome segregation during mitosis. Dysregulation of *CENPA* has been implicated in chromosomal instability and is often associated with malignant transformation and tumor progression.

The presence of these genes at the core of the LUAD PPI network indicates that they likely contribute to central regulatory mechanisms in tumor development, influencing processes such as uncontrolled cell division, genomic instability, and treatment resistance. Their consistent ranking as top hubs across multiple algorithms not only strengthens confidence in their significance but also suggests their potential utility as diagnostic biomarkers or therapeutic intervention points. These findings also align with existing literature that identifies these genes as key players in cancer biology, further validating the computational analysis.

In conclusion, the construction of the LUAD-specific PPI network and subsequent hub gene identification provided a systems-level view of molecular interactions among DEGs. The identification of *CCNB1*, *TOP2A*, and *CENPA* as central hub genes offers promising avenues for targeted research and clinical application, potentially aiding in the development of more effective diagnostic tools and treatment strategies for LUAD.

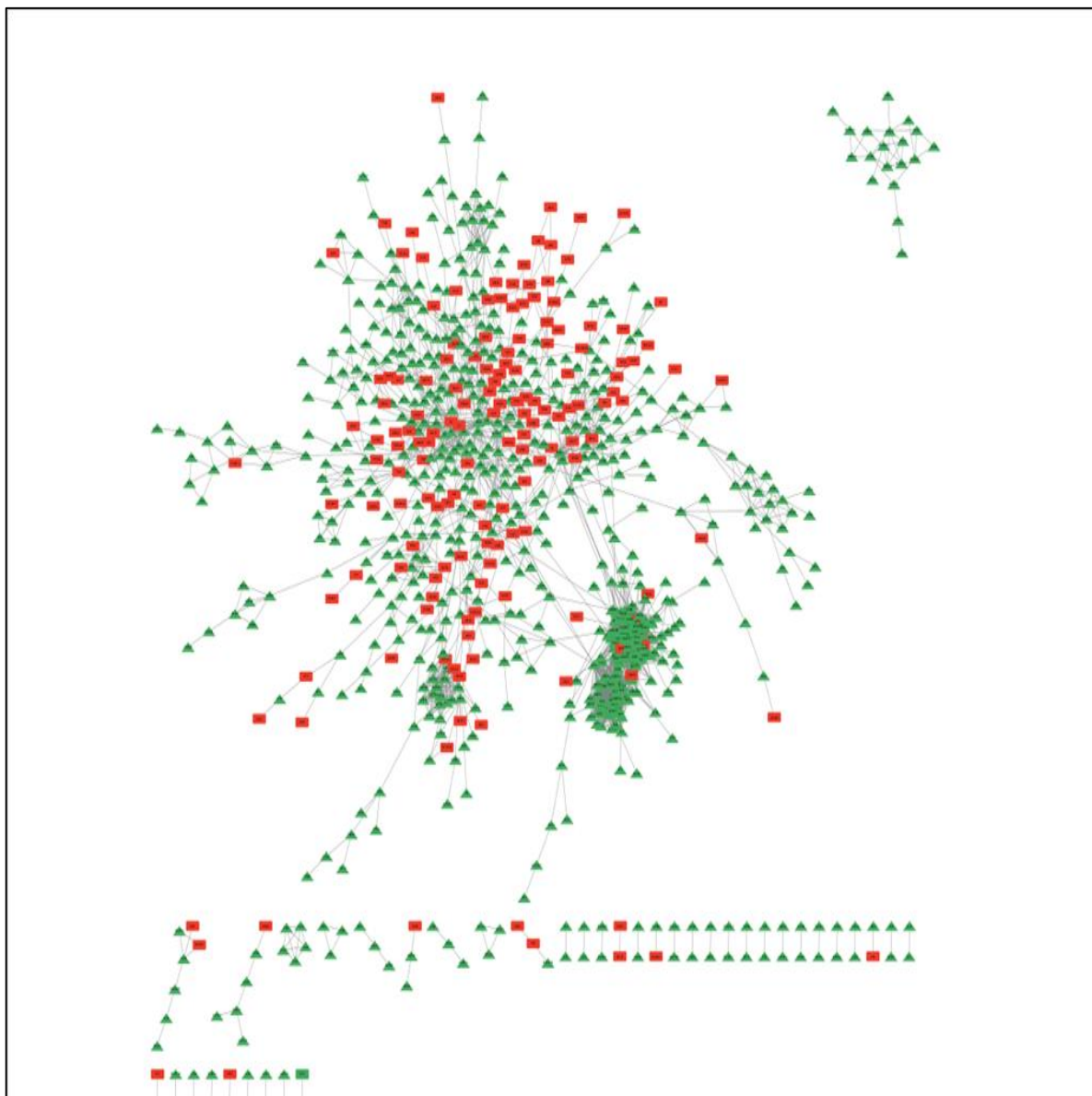
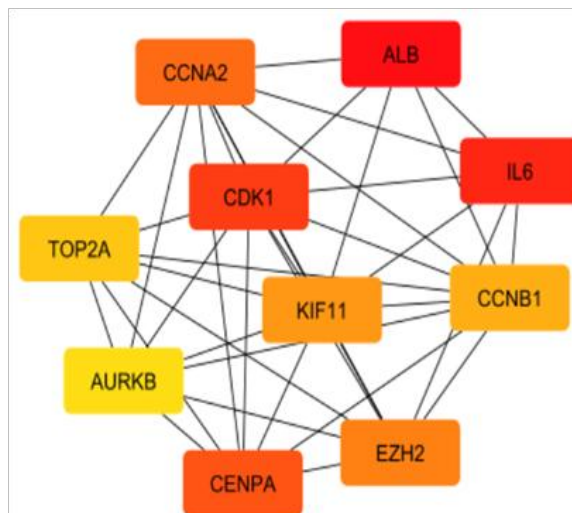
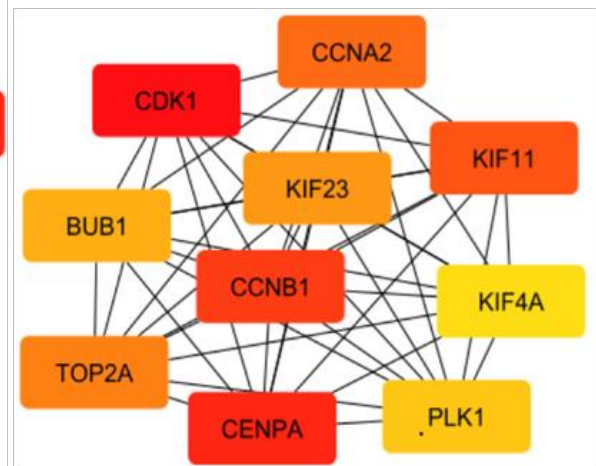


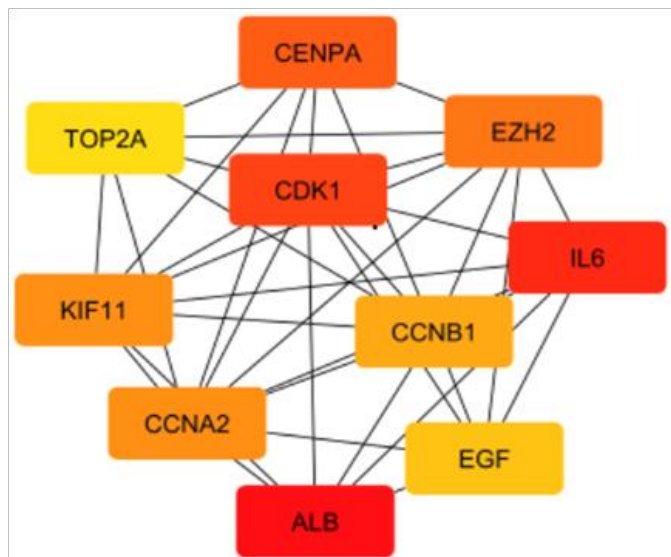
Figure 6.6: Protein–protein interaction network of DEGs. Green triangles indicate upregulated genes; red squares indicate down regulated genes.



MNC method



EPC method



DEGREE method

Figure 6.7 : Hub genes *CCNB1*, *TOP2A*, and *CENPA* were identified using MNC, Degree, and EPC methods.

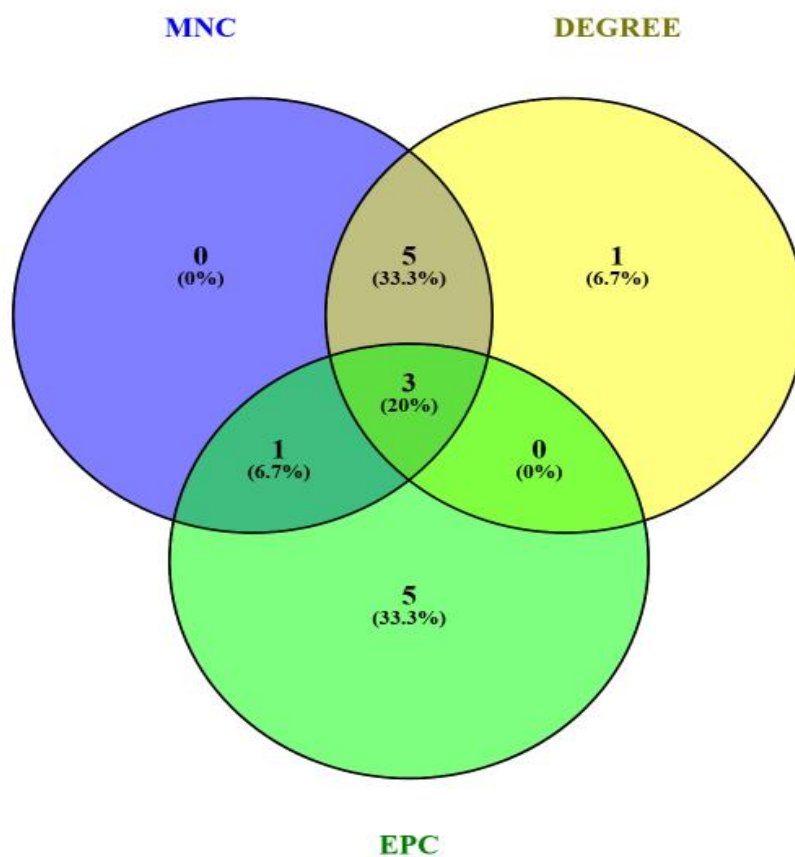


Figure 6.8 : Venn diagram showing the distribution and overlap of hub genes identified through three centrality measures: Maximum Neighborhood Component (MNC), Degree, and Edge Percolated Component (EPC) in lung adenocarcinoma gene network analysis.

Table 6.1: Top Hub genes common in three centralities

S.No	COMMON HUB GENES
1	CCNB1
2	TOP2A
3	CENPA

7.4 Survival Analysis of Hub Genes

To evaluate the prognostic significance of the hub genes identified through the PPI network analysis, *CCNB1*, *CENPA*, and *TOP2A* Kaplan Meier (KM) survival analysis was

conducted. This approach enables assessment of the correlation between gene expression levels and overall survival outcomes in lung adenocarcinoma (LUAD) patients. For each gene, patient samples were stratified into high-expression and low-expression groups based on median expression values, allowing for a direct comparison of survival probabilities over time.

The Kaplan Meier survival curves (Figure 6.9,6.10,6.11) revealed a consistent and significant association between higher expression levels of these hub genes and poorer overall survival among LUAD patients. In all three cases, individuals in the high-expression group (depicted by red lines) showed markedly reduced survival durations compared to those in the low-expression group (depicted by black lines). These differences were not only visually apparent in the KM plots. Still, they were also statistically validated through the log-rank test, which yielded highly significant p-values for each gene, as detailed in Table 6.2.

Moreover, survival analysis yielded elevated hazard ratios (HRs) for all three genes, indicating that patients with increased *CCNB1*, *CENPA*, or *TOP2A* faced a significantly higher mortality risk than those with lower expression. The confidence intervals (CIs) surrounding these hazard ratios further supported the robustness of the findings, demonstrating narrow bounds that excluded unity and thereby confirmed statistical significance. These results highlight a strong and negative prognostic implication of hub gene overexpression in LUAD.

The known functions of the hub genes reinforce the biological plausibility of these observations. As previously discussed, *CCNB1* regulates the G2/M phase transition in the cell cycle, and its dysregulation is commonly associated with unchecked cellular proliferation. Similarly, *TOP2A* is critical in DNA replication and chromosome dynamics, while *CENPA* is essential for centromere function and chromosomal stability during mitosis. Overexpression of these genes reflects heightened proliferative capacity and genomic instability, hallmarks of aggressive tumor behavior that could account for the observed reduction in patient survival.

In summary, survival analysis has established that *CCNB1*, *CENPA*, and *TOP2A* are central to the LUAD interaction network and hold substantial prognostic value. Their high expression correlates strongly with adverse clinical outcomes, suggesting these genes

could be poor prognostic biomarkers. These findings guide future research into stratified therapy approaches or molecular risk profiling in LUAD management.

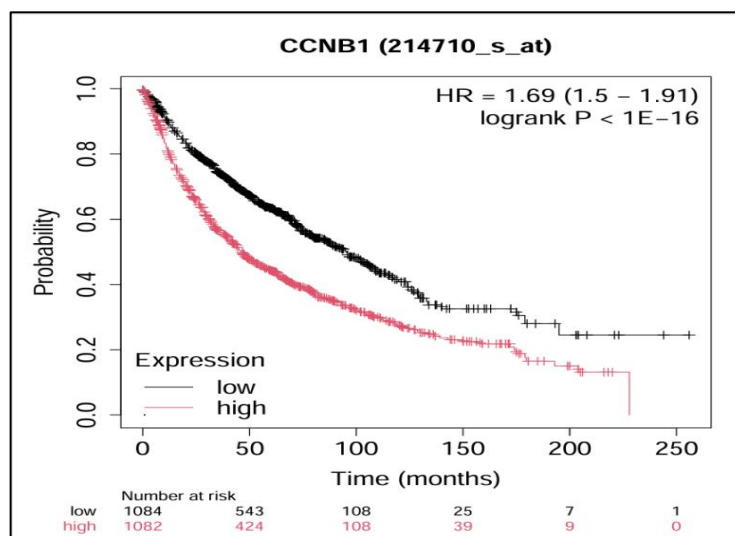


Figure 6.9: Kaplan–Meier survival analysis of LUAD patients stratified by *CCNB1* expression (probe ID: 214710_s_at).

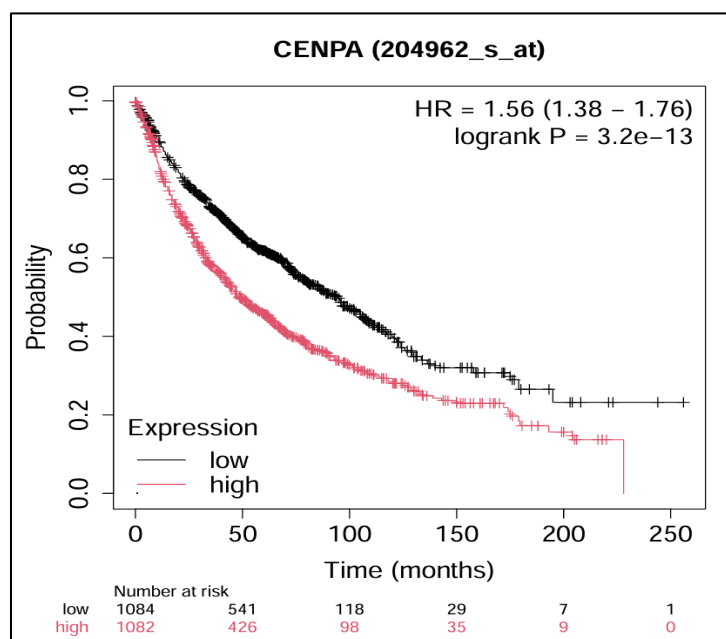


Figure 6.10: Kaplan–Meier survival analysis of LUAD patients stratified by *CENPA* expression (probe ID: 204962_s_at).

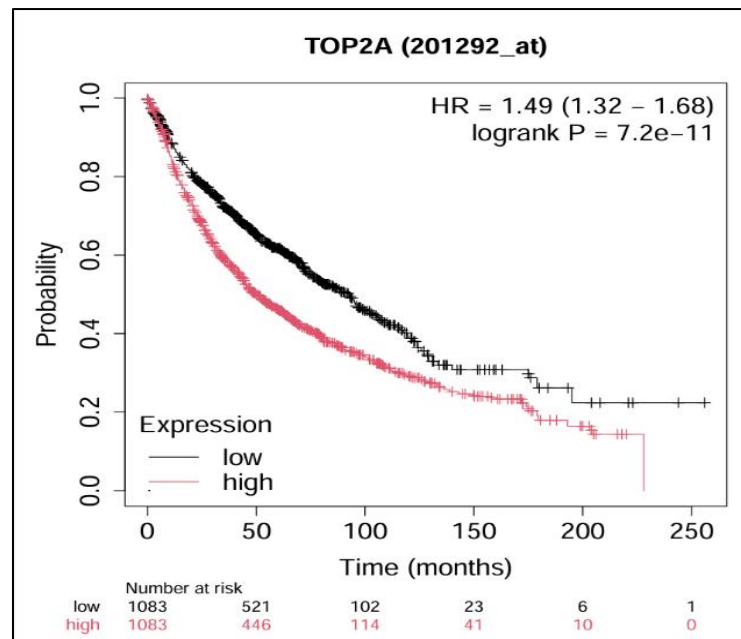


Figure 6.11: Kaplan–Meier survival analysis of LUAD patients based on *TOP2A* expression (probe ID: 201292_at).

Table 6.2: Prognostic significance of candidate hub genes in LUAD based on survival analysis. Hazard ratios (HR), confidence intervals (CI), and log-rank p-values indicate a strong association of high gene expression with poor overall survival.

Gene	Hazard Ratio (HR)	95% Confidence Interval (CI)	Log-rank P-value
CCNB1	1.69	1.5 – 1.91	< 1E-16
CENPA	1.56	1.38 – 1.76	3.2E-13
TOP2A	1.49	1.32 – 1.68	7.2E-11

6.5 ROC-Curve Analysis

To complement the survival analysis and further investigate the potential of the identified hub genes *CCNB1*, *TOP2A*, and *CENPA* as diagnostic biomarkers in lung adenocarcinoma (LUAD), Receiver Operating Characteristic (ROC) curve analysis was performed. This statistical tool enables the evaluation of a gene's ability to distinguish between tumor and normal tissue by plotting the true positive rate (sensitivity) against the false positive rate (1

- specificity) at various threshold settings. The resulting Area Under the Curve (AUC) serves as a summary measure of diagnostic accuracy, with values closer to 1.0 indicating superior discriminatory power. As shown in Figure 6.12, all three hub genes exhibited strong diagnostic capabilities, with ROC curves demonstrating high AUC values, suggestive of robust classification performance in differentiating LUAD from non-cancerous samples. Specifically, the AUC values for *CCNBI*, *TOP2A*, and *CENPA* were consistently high, reflecting reliable predictive strength across gene expression thresholds. This highlights the potential of these genes not only as markers of disease progression but also as early indicators of malignant transformation.

To provide a more nuanced evaluation of diagnostic performance, additional classification metrics were calculated and summarized in Table 6.3. These included Accuracy (Acc), which measures the proportion of correct predictions; Sensitivity (Sens), representing the ability to correctly identify LUAD cases; and Specificity (Spec), assessing the correct identification of non-LUAD samples. The results indicated that all three hub genes achieved high accuracy and sensitivity, with slightly variable but generally acceptable specificity values, underscoring their potential in detecting LUAD with minimal false negatives.

Importantly, the Matthews Correlation Coefficient (MCC) was also reported to assess the overall quality of the binary classifications. MCC is a balanced measure that accounts for true and false positives and negatives and is particularly informative in cases of class imbalance. High MCC scores for *CCNBI*, *CENPA*, and *TOP2A* provide additional confidence in the robustness and reliability of their diagnostic value. Unlike accuracy alone, MCC reflects the consistency of gene performance across both cancer and non-cancer classifications, offering a more comprehensive assessment.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (6.5.1)$$

$$\text{Sensitivity} = \frac{TP}{(TP + FN)} \quad (6.5.2)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (6.5.3)$$

$$\text{MCC} = \frac{(TP * TN - FP * FN)}{\sqrt{((TP + FP) * (TP + FN) * (TN + FP) * (TN + FN))}} \quad (6.5.4)$$

The high diagnostic performance of these hub genes is biologically consistent with their known roles in cell cycle regulation, chromosomal maintenance, and proliferation. The dysregulation of such essential genes is expected to result in significant transcriptomic alterations, which, in turn, make them excellent candidates for biomarker development. Their combined utility in both prognosis (as demonstrated through survival analysis) and diagnosis (as evidenced by ROC curve performance) strengthens the case for their integration into clinical workflows, either as part of a gene panel or as individual targets for molecular testing. In conclusion, ROC curve analysis has demonstrated that *CCNB1*, *CENPA*, and *TOP2A* possess strong diagnostic value, with high AUC scores and favorable classification metrics indicating their ability to reliably distinguish LUAD samples from normal controls. These results support the broader utility of these hub genes as dual-purpose biomarkers, offering both diagnostic and prognostic insights in the clinical management of LUAD.

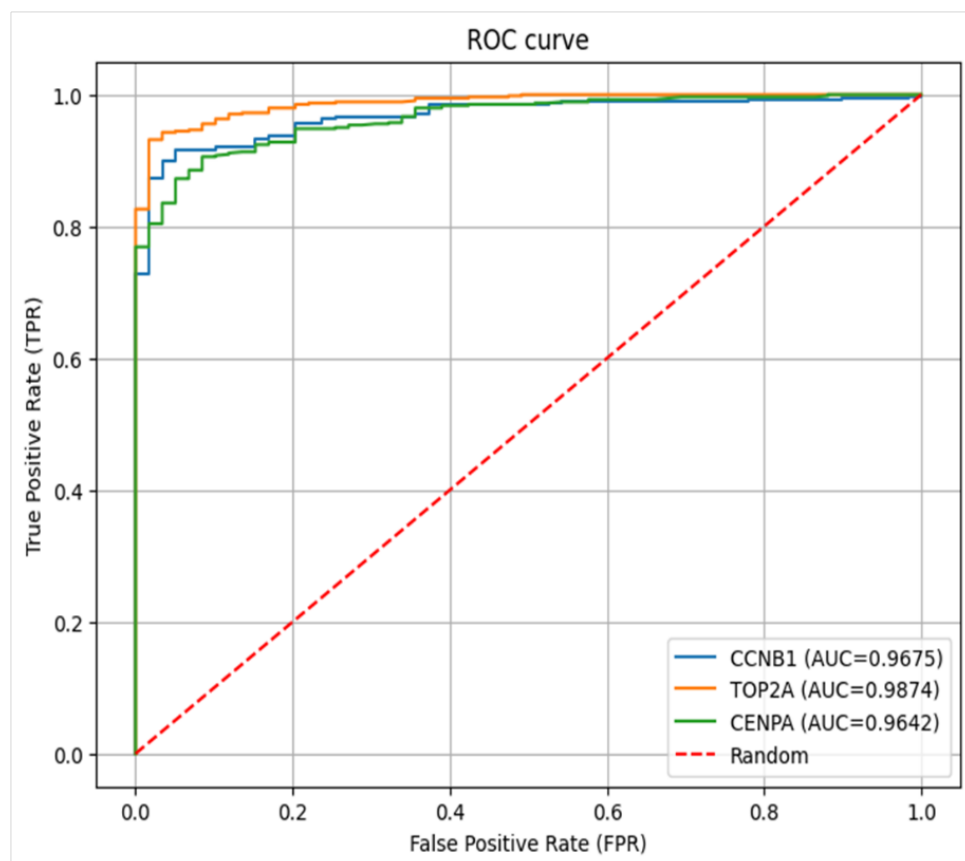


Figure 6.12: ROC curves illustrating the diagnostic performance of *CCNB1*, *CENPA*, and *TOP2A* in classifying LUAD samples.

Table 6.3: Classification performance metrics (Accuracy, Sensitivity, Specificity, MCC) of top hub genes in LUAD.

Gene	Acc	Sens	Spec	MCC
CCNB1	85.83	84.46	98.33	0.59
TOP2A	84.83	83.35	98.33	0.58
CENPA	80.00	77.99	98.33	0.51

6.6 Expression of Hub Genes Across LUAD Stages

To further explore the clinical relevance and dynamic expression patterns of the identified hub genes *CCNB1*, *CENPA*, and *TOP2A* their expression profiles were analyzed across different pathological stages of lung adenocarcinoma (LUAD) using transcripts per million (TPM)-normalized RNA-seq data. This analysis aimed to assess whether the expression of these genes correlates with LUAD progression, providing additional insights into their potential roles in tumor development and stage-specific behavior.

As illustrated in Figure 6.13,6.14,6.15, the expression patterns of *CCNB1* and *TOP2A* demonstrated a clear and progressive upregulation from Stage I to Stage IV, indicating a potential involvement in tumor advancement and aggressiveness. This gradual increase in expression across advancing stages suggests that these genes may not only contribute to early tumor initiation but also play critical roles in sustaining tumor proliferation, genomic instability, and metastasis as the disease worsens. In particular, the stepwise elevation in expression aligns with known mechanisms of tumor evolution, where enhanced activity of cell cycle and replication-related genes becomes increasingly pronounced in later stages to support rapid cancer cell division and survival under stress conditions.

CENPA, while also significantly overexpressed in LUAD tissues compared to normal lung samples, exhibited a more variable pattern across stages. Although its expression was elevated overall, fluctuations in *CENPA* levels between stages suggest that its role may be more pronounced during early or intermediate tumor development, or potentially influenced by other factors such as tumor heterogeneity or subtype-specific regulation. Nevertheless, the consistent overexpression of *CENPA* in cancerous versus non-cancerous

tissue still reinforces its significance as a potential tumor marker, even if its expression trend across stages is less linear than *CCNBI* and *TOP2A*.

These expression trends collectively reinforce the functional importance of these hub genes in LUAD biology. The progressive upregulation of *CCNBI* and *TOP2A* highlights their likely contribution to tumor aggressiveness, potentially through mechanisms involving cell cycle dysregulation, DNA replication stress, and chromosome segregation errors all of which are hallmarks of advanced malignancies. Their rising expression levels with advancing tumor stages may also reflect increased demands on proliferative machinery and may serve as indicators of tumor burden or progression.

Furthermore, this stage-specific expression analysis provides important translational value. Genes that show strong positive correlations with tumor stage can serve as stage-specific biomarkers or inform prognostic models to better stratify patients based on molecular characteristics. In clinical practice, monitoring such genes might enable oncologists to estimate tumor aggressiveness or predict potential disease trajectories, contributing to personalized treatment planning.

In conclusion, expression profiling of *CCNBI*, *CENPA*, and *TOP2A* across LUAD stages reveals significant and biologically meaningful trends. The stage-wise increase in *CCNBI* and *TOP2A* supports their role in LUAD progression and highlights their potential utility as molecular markers for tumor staging. Although *CENPA* exhibited more variable expression, its consistently elevated levels in tumor tissues still affirm its relevance in LUAD pathophysiology. Together, these findings underscore the diagnostic, prognostic, and staging-related utility of these hub genes in lung adenocarcinoma and provide a foundation for their further investigation in clinical and therapeutic contexts.

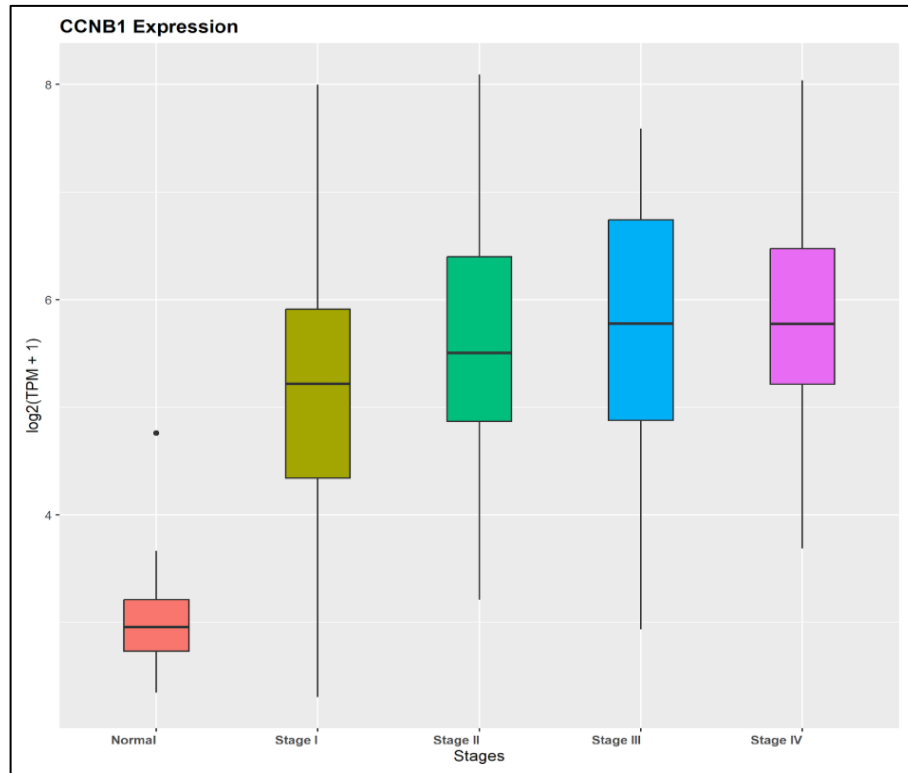


Figure 6.13: Boxplots representing expression levels of *CCNB1* across normal lung tissue and LUAD stages (I–IV).

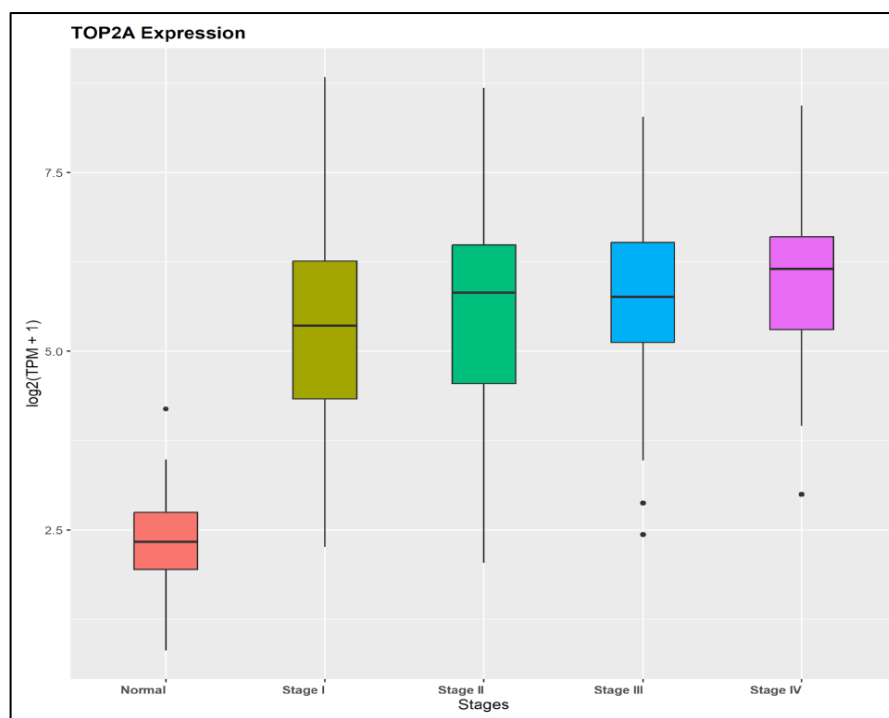


Figure 6.14: Boxplots representing expression levels of *TOP2A* across normal lung tissue and LUAD stages (I–IV).

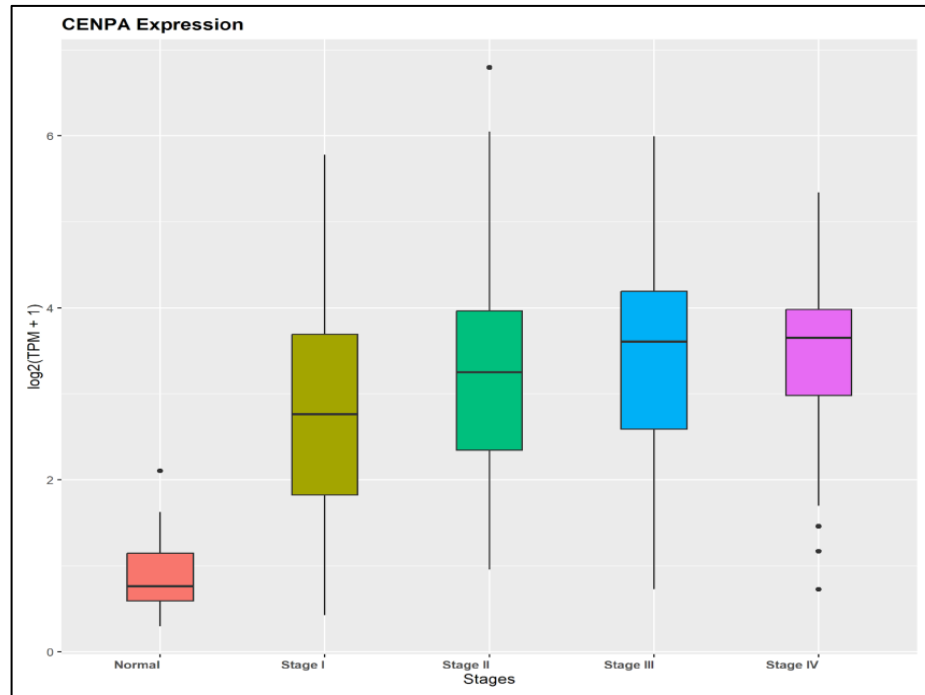


Figure 6.15: Boxplots representing expression levels of *CENPA* across normal lung tissue and LUAD stages (I–IV).

6.7 Fairness Evaluation analysis

To assess potential bias in our machine learning model, we conducted a fairness evaluation by comparing model performance across gender groups under three different dataset configurations: (1) the original dataset, (2) a class-balanced dataset using SMOTE, and (3) a class and gender balanced dataset using SMOTE followed by stratified random gender sampling.

Figure 6.16. shows the initial imbalance in class distribution (Tumor vs Normal) and gender (Male vs Female), highlighting that male and female patients were not equally represented across classes. This imbalance can introduce unintended bias, particularly in gender-sensitive clinical applications. To address this, we applied synthetic oversampling (SMOTE) to balance the classes and combined it with random gender-based stratification to balance the genders. This resulted in a more equitable representation of both males and females in the final dataset.

Demographic Parity Evaluation:Demographic parity measures whether each demographic group (here, gender) receives positive predictions at equal rates, regardless of their true labels. It is calculated as:

$$\text{Demographic Parity} = \text{TP} + \text{FP} \quad (6.7.1)$$

As shown in Figure 6.17, the original dataset had a significant disparity in demographic parity between females and males. While class balancing alone slightly improved parity for the female group, the combined class and gender balancing approach achieved the most equitable distribution, indicating a reduced disparity between genders in model outcomes.

Accuracy Parity Evaluation:Accuracy parity assesses whether model accuracy is consistent across demographic groups. It is calculated as:

$$\text{Accuracy Parity} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{TN} + \text{FN}) \quad (6.7.2)$$

In the original dataset, model accuracy for males (0.60) significantly outperformed that for females (0.51), demonstrating a clear gender bias. With class balancing, female accuracy improved to 0.78, though male accuracy dropped to 0.70. However, the best outcome was achieved under the class and gender-balanced setup, where female and male accuracies were nearly equal (0.74 and 0.75 respectively), as visualized in Figure 6.18.

These findings highlight the importance of addressing both class and gender imbalances in biomedical datasets to achieve fair and reliable model performance. In clinical settings where equitable treatment is critical, such fairness-aware approaches help ensure that predictions are not skewed due to bias.

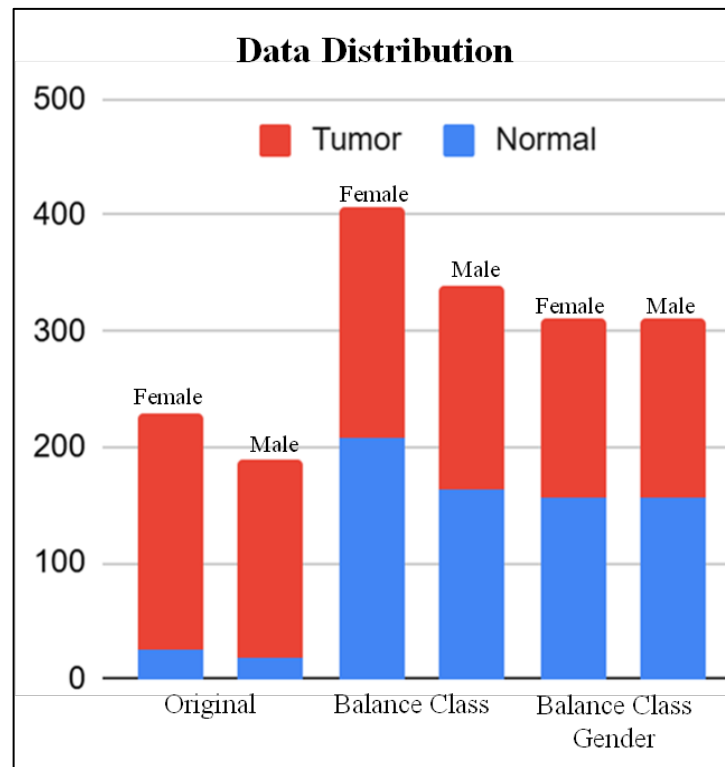


Figure 6.16: Data distribution plot showing the number of tumor and normal cases by gender across original, class-balanced, and class+gender-balanced datasets.

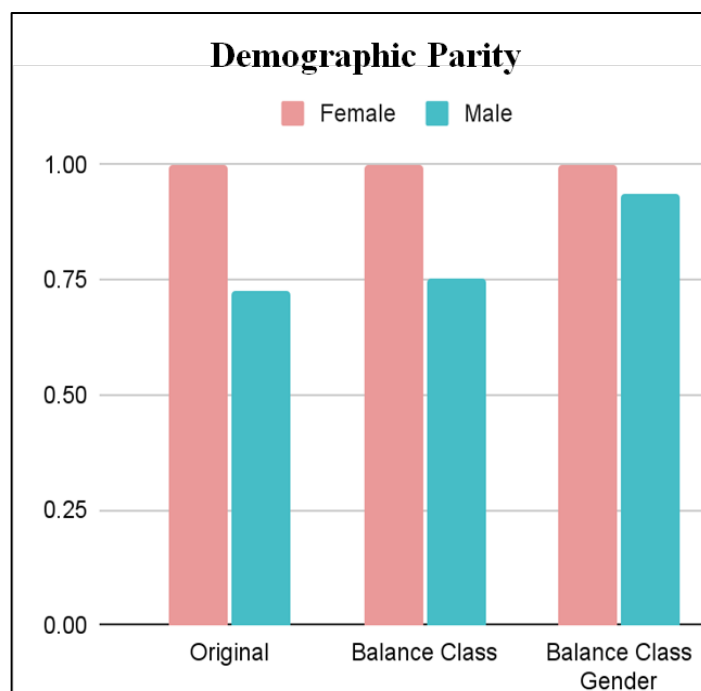


Figure 6.17: Bar plot showing demographic parity scores for male and female groups across original, class-balanced, and class+gender-balanced datasets.

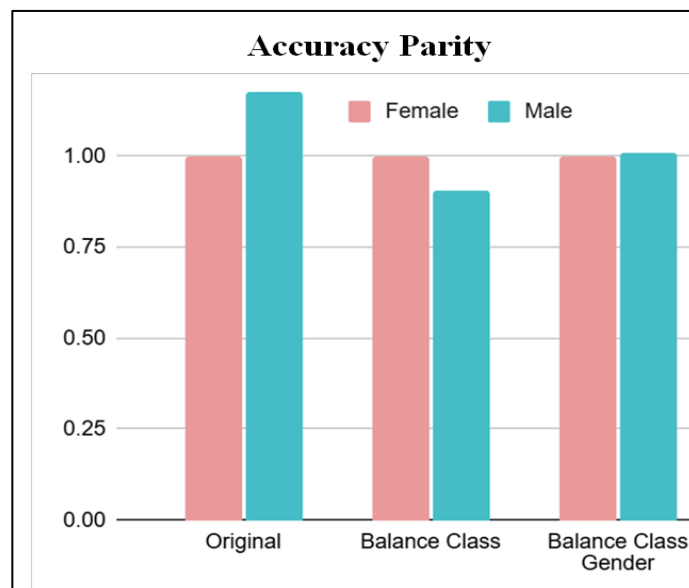


Figure 6.18: Accuracy parity across gender groups in original, class-balanced, and class and genderbalanced datasets.

Table 6.4: Comparison of machine learning model accuracy across gender groups using different balancing approaches for lung adenocarcinoma prediction

Accuracy		
Data	Female	Male
Original	0.51	0.6
Balance Class	0.78	0.7
Balance Class Gender	0.74	0.75

6.8 ROC curve analysis for the fair model

The ROC curve analysis for the selected fair model demonstrates excellent diagnostic performance across three key biomarkers for lung adenocarcinoma detection. TOP2A demonstrated superior performance across all metrics, with the highest AUC (0.9886), accuracy (91.33%), specificity (99.33%), and Matthews correlation coefficient (0.8382). CCNB1 and CENPA also showed robust discriminatory power with AUCs of 0.9705 and 0.9651, respectively. While all three biomarkers exhibited excellent specificity (>98%), their relatively lower sensitivity values (77.56-83.56%) suggest they excel at ruling out disease in healthy individuals but might benefit from complementary markers to improve detection in true positive cases. Importantly, our gender-balanced fair model successfully addressed previously identified biases, producing a more equitable diagnostic tool without compromising performance, as evidenced by the consistently high AUC and MCC values across all biomarkers. These findings validate these genes as reliable clinical biomarkers with significant potential for improving lung adenocarcinoma diagnosis.

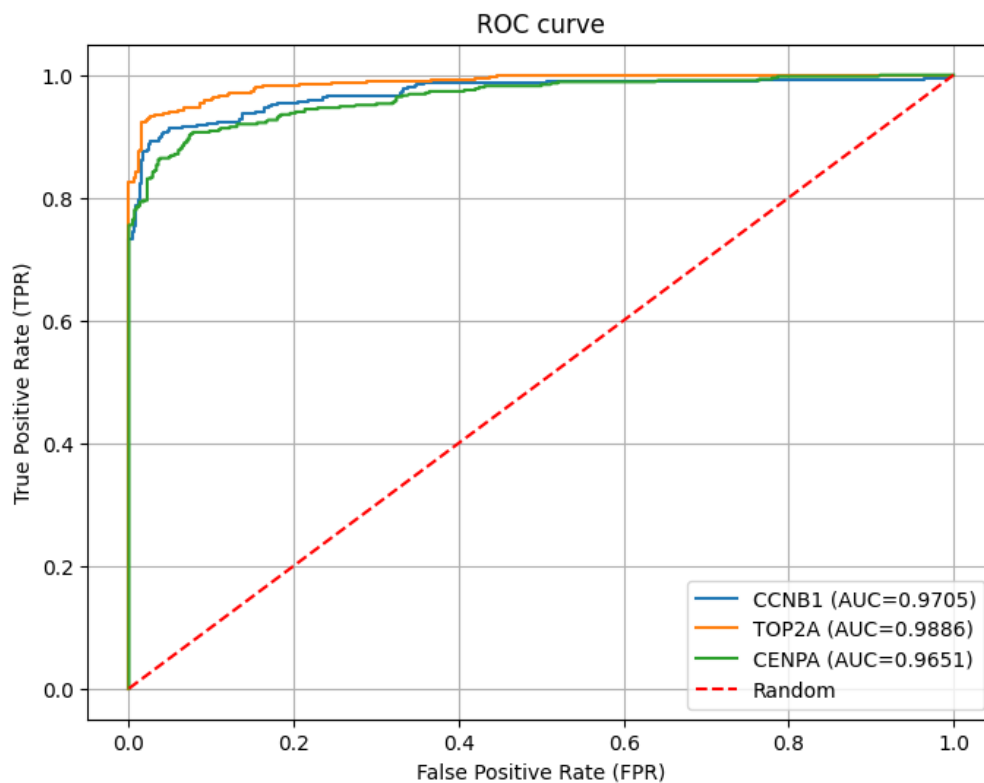


Figure 6.19 : ROC curves (Logistic Regression with CV-10) and performance metrics of fair models for hub genes after class and gender balancing

Table 6.5: Performance metrics of identified biomarkers *CCNB1*, *TOP2A*, and *CENPA* in lung adenocarcinoma prediction, showing accuracy, sensitivity, specificity, and Matthews correlation coefficient values.

Gene	Acc	Sens	Spec	MCC
CCNB1	91.00	83.56	98.44	0.8301
TOP2A	91.33	83.33	99.33	0.8382
CENPA	88.44	77.56	99.33	0.7886

7. DISCUSSION

The most common subtype of non-small cell lung cancer, named lung adenocarcinoma (LUAD), is responsible for a significant percentage of cancer-related mortality globally due to its aggressive nature and late stage of diagnosis [2]. The limitation of identifying reliable molecular biomarkers for earlier detection and therapy choice has fueled increased research into transcriptomic-based approaches. In this research, an extensive RNA-Seq analysis of LUAD samples resulted in the detection of 2,269 differentially expressed genes (DEGs), and among them, three hub genes *CCNB1*, *CENPA*, and *TOP2A* were identified as major regulators based on network structure, expression, and clinical significance.

These genes play central roles in cell cycle control, fidelity of mitosis, and chromosome stability. *CCNB1*, encoding Cyclin B1, is a recognized cell cycle regulator that facilitates G2/M transition and is often overexpressed in virulent tumors, such as LUAD, to drive uncontrolled proliferation and unfavorable prognosis [12,14]. *CENPA*, a histone H3 variant that plays a crucial role in kinetochore assembly and centromere identity, has been linked to tumor cell plasticity, stemness, and epithelial–mesenchymal transition, particularly in highly proliferating cancers [64]. *TOP2A*, a DNA topoisomerase, participates in DNA unwinding and is a classic target of anti-cancer drugs; its overexpression has also been linked to poor prognosis in several cancer types, such as hepatocellular and lung cancers[65].

The expression of these hub genes by expression profiling showed strong overexpression in tumor tissues compared to normal tissue, with increasing levels in increasingly advanced cancer stages. These patterns indicate a potential function in tumor progression and are excellent candidates for diagnostic and prognostic biomarkers. Survival analysis with the KM Plotter tool validated that increased expression of these genes is associated with highly significant decreased overall survival, which again underscores their clinical usefulness. High diagnostic accuracy by ROC curve analysis also supports their use as molecular classifiers, with AUC > 0.96 for all three genes[66].

Functional annotation and pathway analysis of DEGs indicated that they were engaged in major cancer-related processes, including mitotic spindle assembly, DNA repair, hormone signaling, and immune regulation, biological themes frequently impaired in lung adenocarcinoma. Remarkably, the results concordant with previous research that reported comparable functional patterns in LUAD based on multi-omics and immune infiltration analyses[67,68].

A new feature of this research is the incorporation of fairness assessment into machine learning models applied to gene-based classification. Machine learning software is increasingly used in clinical genomics; however, dataset biases can lead to disparate model performance across demographic subgroups, which can cause potential healthcare disparities. To mitigate this, fairness metrics, such as demographic and accuracy parity, were employed to assess predictive equity between gender groups. By using SMOTE to balance both class and gender distributions, the models achieved better fairness and accuracy. This is based on more recent research that promotes transparent and ethical AI in health systems [71,72,73]

Despite this, the study is not without limitations. As a computational study, it depends mainly on in silico predictions and publicly available datasets. Hence, experimental verification is necessary to establish the biological functions and regulatory processes of CCNB1, CENPA, and TOP2A in LUAD tumorigenesis. Gene knockdown, overexpression, and protein interaction experiments may yield more mechanistic insights in lung cancer cell lines. Further, testing the therapeutic response related to modulation of these genes may reveal new drug targets or markers of resistance. In addition, broadening fairness evaluation to other demographic factors such as age, ethnicity, and clinical site would enhance the generalizability and ethical strength of predictive models in practical use.

This research identifies three biologically meaningful and clinically relevant hub genes that may be used as multi-purpose biomarkers for LUAD, and further discusses fairness in cancer diagnosis. By integrating transcriptomic discovery and fairness in machine learning, we take one step closer to more personalized, precise, and fair cancer diagnosis.

8. CONCLUSION

This study adopted a comprehensive approach to identify and validate robust biomarkers for lung adenocarcinoma (LUAD) by integrating transcriptomic analysis, network biology, machine learning, and fairness-aware strategies. Analysis of RNA-Seq data from The Cancer Genome Atlas (TCGA) identified 2,269 differentially expressed genes, reflecting molecular alterations associated with LUAD progression. Functional enrichment highlighted the involvement of these genes in key biological pathways, including cell cycle regulation, hormone signaling, and metabolic processes central mechanisms in cancer development. Protein-protein interaction network analysis revealed three consistently significant hub genes: *CCNB1*, *TOP2A*, and *CENPA*. These genes demonstrated strong prognostic value, with Kaplan-Meier survival analysis indicating that elevated expression levels correspond to reduced overall survival. Receiver Operating Characteristic (ROC) curve analysis further confirmed their diagnostic potential, with area under the curve (AUC) values exceeding 0.96. An essential component of the study was the implementation of fairness-aware machine learning to examine ethical considerations in predictive modeling. Models trained on datasets balanced for class and gender achieved enhanced fairness metrics while maintaining high predictive accuracy. The final model demonstrated comparable performance across male and female subgroups, underscoring its potential for equitable clinical application. Overall, the findings identify clinically relevant biomarkers and introduce a framework for integrating fairness into cancer diagnostics. The methods and insights from this study lay a solid foundation for developing equitable precision medicine strategies, ensuring that our predictions are accurate and inclusive for all patients.

9. REFERENCES

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